Reconciling Efficiency and Effectiveness of Exercise Retrieval: An Uncertainty Reduction Hashing Approach for Computerized Adaptive Testing

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Abstract

With the rapid development of intelligent education, Computerized Adaptive Testing(CAT) has garnered significant attention for its ability to tailor exercises to individual examinees. The adaptability of CAT is primarily achieved through the alternating optimization of two core components: the cognitive diagnosis model and the exercise selection module. However, existing CAT approaches, despite their remarkable achievements, often come at the expense of high time costs. Statistical-based approaches incur increased time overhead due to complex computations, while data-driven approaches further exacerbate time inefficiency because of the iterative processes in reinforcement learning, making it challenging to balance evaluation effectiveness and time efficiency. To this end, in this paper, we propose **HashCAT**, an efficient CAT approach based on learning to hash, aiming to balance efficiency and evaluation effectiveness. Our approach comprises two stages: the hash representation generation and the exercise selection. In the first stage, we design an information alignment module and a novel cognitive diagnosis function to model the interaction between examinees and exercises, generating hash representations with clear physical significance. In the second stage, we propose an uncertainty reduction-based algorithm that utilize information entropy to quantify the uncertainty in student ability estimation and selects exercises that most effectively reduce this uncertainty. Experimental results on four real-world datasets demonstrate that the proposed method significantly improves question selection efficiency while maintaining competitive evaluation performance. The code exists anonymously in https://github.com/sherklock/Intelligent-Education/tree/main/HashCAT-main.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction**.

Keywords

Computerized Adaptive Testing, Exercise Retrieval, Learning to hash

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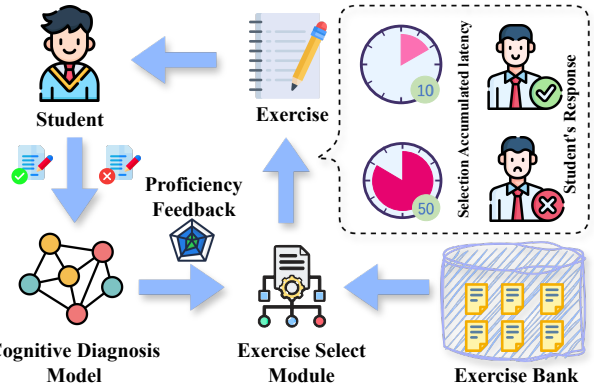


Figure 1: Illustration of a typical CAT process, highlighting time efficiency as a critical issue in exercise retrieval.

1 Introduction

With the rise of personalized learning [6, 41, 44], computerized adaptive testing (CAT) [30] has garnered significant attention as a promising ability assessment technique that aims to effectively evaluate learners' mastery of specific knowledge concepts by incorporating pedagogical theories. Unlike traditional paper-and-pencil tests, the unique adaptive nature of CAT customizes questions for each individual, accurately providing proficiency portrayal of the learner, often described as "tailored to each examinee" [23, 31].

In general, the adaptivity of CAT is achieved through the inclusion of two alternately optimized core components [42]: a cognitive diagnosis model (CDM) and an exercise selection module (ESM). As illustrated in Figure 1, the CDM [7, 18, 21, 39, 43] aims to portray the examinee's current knowledge level by analyzing his/her previous responses to exercises, and based on the acquired proficiency, the ESM personalizes the selection of suitable and effective questions to assist in the next step of the measurement. The two assisted each other, alternately repeating until a pre-determined stopping condition is met, at which point the learner's final proficiency estimate is recorded as the definitive result of the CAT process.

In recent years, CAT has developed extensively. Researchers have primarily focused on enhancing its ability to evaluate competencies by introducing advanced neural network-based cognitive diagnosis models and improving exercise selection strategies. In general, exercise selection algorithms in existing CAT approaches can be categorized into two types: statistical-based approaches [1, 2, 15, 51] and data-driven approaches [8, 34, 46, 50]. Specifically, statistical-based approaches optimize the adaptive evaluation process by assigning a value (e.g., Fisher Information) to each exercise and selecting the one that maximizes the information gain based on the current estimate of the learner's ability [14, 17, 47]. For example, MAAT [1] consists of three key modules—the quality, diversity, and importance modules—that work together to evaluate the information content of each exercise by assessing the expected model change. In comparison, data-driven approaches automatically learn and optimize the exercise selection algorithm based on learner's previous interaction records [14], thus avoiding the complexity associated with designing statistical-based approaches. For instance, NCAT [50] formulates CAT as a bi-level optimization problem, addresses it as an equivalent reinforcement learning task, and proposes an attentive neural policy to model complex non-linear interactions.

However, whether statistical-based or data-driven approaches, while achieving remarkable performance, these accomplishments come at the cost of significant time consumption. In particular, existing statistical-based approaches involve complex computations when calculating the information content of exercises, leading to a significant increase in time costs when handling large exercise banks. In contrast, data-driven approaches optimize the exercise selection algorithm using reinforcement learning strategies, and due to the agent-environment interaction and iterative process, they also face high time overhead [14]. Given this, the study aims to optimize the balance between time efficiency and evaluation effectiveness in CAT exercise selection algorithms. Inspired by previous research [25], we conceptualize CAT as an exercise retrieval task. Unlike traditional CAT approaches, the information retrieval-based selection process is viewed as a matching process between the examinee and exercises, improving selection efficiency through fast retrieval by utilizing existing information. Building on this, we introduce the concept of learning to hash, employing discrete hash representations for both examinees and exercises, and utilizing the low computational complexity of hash representations to improve retrieval efficiency.

However, implementing this concept in CAT presents several significant challenges. To fully realize the potential of learning to hash in this context, some critical questions must be addressed. (1) how can the hash representations of exercises and examinees be effectively generated? In the CAT context, we need to obtain hash representations with clear physical meaning. (2) how can the hash representation generation process be optimized? Traditional approaches for generating hash representations [29] are insufficient to fully capture the complex interactive relationship between learners and exercises. For the discrete representations of examinees and exercises, we need to design a new diagnostic function to accurately describe the interaction between them, thereby optimizing the hash representation generation mechanism. (3) how can an effective exercise selection strategy be designed based on hash representations? We need to design a reliable exercise selection algorithm that fully utilize the computational properties of hash representations to enhance retrieval efficiency, while effectively avoiding a decline in the reliability of assessment results due to information loss.

To this end, in this paper, we propose **HashCAT**, a CAT approach based on uncertainty reduction and learning to hash, which comprises two stages: hash representation generation and exercise selection process. Specifically, in the hash representation generation stage, we obtain the hash representations of learners and exercises based on their latent embeddings. In order to ensure that the hash representation and the real-valued representation are as consistent as possible in semantic information and structure, we designed an information unification module. Additionally, to optimize the hash representation generation process in the CAT scenario, we have proposed a novel diagnostic function that captures the interactive relationship between learners and exercises, thereby optimizing the hash representation generation process. In the exercise selection stage, we design an exercise selection algorithm based on uncertainty reduction, utilizing the obtained hash representations and exercise features. We quantify the uncertainty in student ability assessment using information entropy and select exercises that most effectively reduce this uncertainty. Finally, we validated our

approach on four real-world datasets, demonstrating its balance between time efficiency and evaluation effectiveness.

2 Related Work

2.1 Computerized Adaptive Testing

Computerized Adaptive Testing (CAT) improves the accuracy of assessing examinees' abilities by alternating between a cognitive diagnosis model and a exercise selection algorithm, utilizing a limited number of item selection steps [19].

The exercise selection algorithm in CAT has long been a central focus of research. In traditional CAT methods, two statistical-based selection algorithms, Maximum Fisher Information [16] and Kullback-Leibler Information [2], are the most widely used, given the characteristics of Item Response Theory (IRT) [24]. BECAT [51], in contrast, finds an approximate estimate of the examinee's true ability through theoretical estimation, and based on this, proposes a bounded ability estimation framework for CAT. Subsequently, data-driven approaches have gained widespread attention [9, 34, 50]. A representative approach is GMOCAT [34], which describes the relationship between exercises and knowledge concepts using graph neural networks [38, 45]. Besides, to address the issue where the time complexity of existing topic selection strategies is linearly related to the number of exercises, SECAT [10] introduces a CAT framework with efficient search capabilities. Although this framework can improve exercise selection efficiency, the preliminary work it requires is complex and time-consuming. These works almost all focus on improving evaluation effectiveness but rarely address the key issue of exercise selection efficiency. Thus, our goal is to design a new CAT approach that enhances time efficiency while maintaining evaluation effectiveness.

2.2 Learning to Hash

In recent years, the explosive growth of data has sparked significant interest in efficient retrieval methods. Among mainstream search methods, learning to hash has gained widespread attention due to its ability to utilize the inherent properties of data. The primary goal of learning to hash is to design task-specific hash functions that map raw data into compact binary codes, thereby enhancing search efficiency while preserving retrieval accuracy [37]. With the rise of deep learning, which excels at feature representation, deep learning-based hashing methods have demonstrated impressive performance across various domains. Recent advancements in these methods—such as deep supervised hashing, triplet loss functions, and neural network-based quantization, have further advanced both retrieval effectiveness and efficiency [26].

In the CAT, despite the widespread use of learning to hash, there are still three main challenges: obtaining physically meaningful hash representations of examinees' abilities and exercise difficulties, designing a diagnostic function based on hash representations and optimizing hash generation, and creating efficient exercise selection strategies that balance time efficiency and assessment accuracy.

3 Preliminaries

In a conventional intelligent education scenario, CAT can adaptively select exercises for examinees based on their proficiency feedback, to accurately assess their abilities. In CAT, the examinee population is represented by the set $\mathcal{S} = \{s_1, s_2, \ldots, s_{|\mathcal{S}|}\}$, and the exercise bank is represented by the set $\mathcal{E} = \{e_1, e_2, \ldots, e_{|\mathcal{E}|}\}$, the knowledge concepts is represented by the set $\mathcal{C} = \{k_1, k_2, \ldots, k_{|C|}\}$. A examinee's practice log $\mathcal{R}_s$ is generally defined as a triplet: $\mathcal{R}_s(s_i, e_j, r_{ij})$, where if examinee s_i answers exercise e_j correctly, r_{ij}=1, otherwise r_{ij}=0, with $s_i \in \mathcal{S}$ and $e_j \in \mathcal{E}$. In the CAT system, it is assumed that the true ability of a examinee does not change. A classic CAT system consists of two core modules: the cognitive diagnosis module $\mathcal{M}$ and the exercise selection algorithm π [19]. The CAT process can be represented as:

$$\begin{aligned} e_i^t &\leftarrow \pi(\theta_i^{t-1} | Q_u, \Omega), \\ \theta_i^t &\leftarrow \mathcal{M}(e_i^t | \xi), \end{aligned} \tag{1}$$

where Q_u denotes the candidate exercise bank for examinee s_i, θ_i^{t-1} denotes the proficiency feedback of the examinee s_i at test step $t-1$, e_i^t denotes the untested exercise selected for examinee s_i at test step t by the exercise selection algorithm. Additionally, θ_i^t denotes the proficiency of examinee s_i assessed by $\mathcal{M}$ at test step t. Ω, ξ are the parameters of $\mathcal{M}$ and π, respectively. In CAT, the process stops after $\mathcal{M}$ and π alternate for T steps, where T is the maximum number of steps for terminating the test.

In our work, we input examinee's historical learning logs $\mathcal{R}$ into our designed hash representation extraction model, generating the hash representation of examinee's proficiency b_s and the hash representation of exercise's difficulty b_e, along with the real-valued difficulty and discrimination of the exercises. We also propose a novel diagnostic function specifically designed for hash representations to model the interaction between examinees and exercises, thereby refining the hash representation generation module. Our hash-based representation learning extracts multi-dimensional exercise features. The selection algorithm integrates examinee proficiency with exercise characteristics, prioritizing exercises that minimize ability uncertainty through entropy-driven criteria. This framework ensures reliable assessment while achieving sublinear-time retrieval efficiency.

4 Methodology

Overview. Our HashCAT framework enhances CAT efficiency by leveraging hash representations for examinees and exercises. As illustrated in the Figure 2, it comprises four components: (1) Hash Representation Generation Module (HRGM), (2) Information Alignment Module (IAM), (3) Hash-based Diagnosis Module (HDM), and (4) Uncertainty Reduction-based Exercise Selection. Specifically, the HRGM is responsible for generating hash representations of the knowledge concepts proficiency vectors of the examinees and the hash representations of exercise difficulty. But the process of converting real-valued representations into hash representations is well know to suffer from quantization loss [36], leading to significant discrepancies between the two representations. To address this, the IAM is designed to minimize information loss during the generation of hash representations, ensuring that the hash representations retain as much information as possible from their real-valued counterparts. Besides, to accurately describe the interactive relationship between examinees and exercises under the hash representation framework, we introduce a novel interaction function in the HDM. This function is specifically designed to optimize the hash representation generation process. Finally, by utilizing the hash

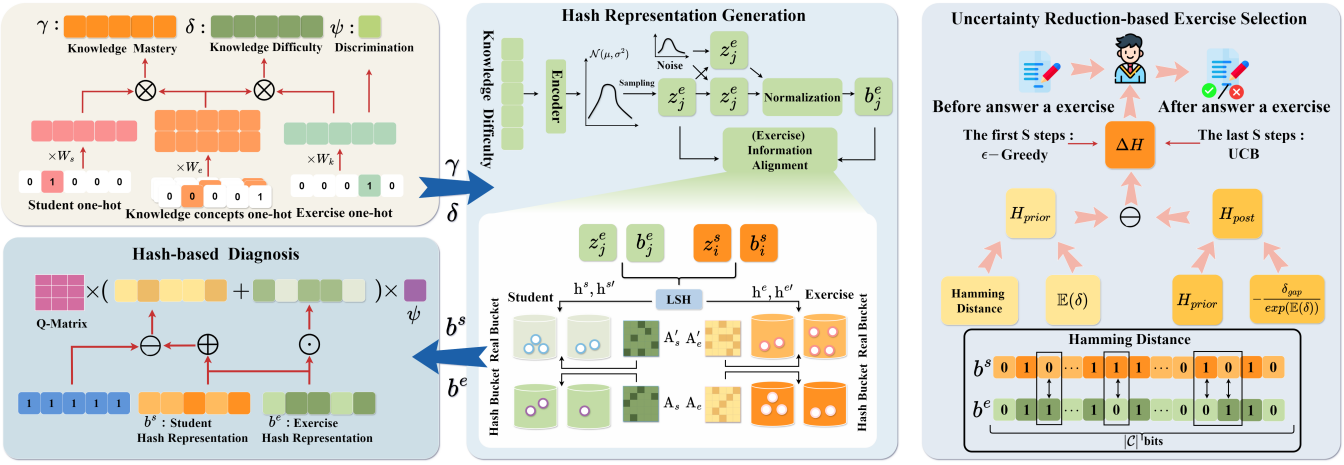


Figure 2: Illustration of our HashCAT framework. (a) and (b) together form the hash representation generation process, which includes the HRGM, IAM, and HDM modules. (c) represents the Uncertainty Reduction-based Exercise Selection.

representations of examinees and exercises, we perform the actual CAT exercise selection process under the Uncertainty Reduction Strategy, ensuring efficiency and evaluation effectiveness. In each step of the CAT process, the real-valued representation of examinee proficiency is provided by $\mathcal{M}$, while our hash representation generation module produces the hash representation of exercises. During the selection process, we apply the same generation strategy to the real-valued representation of examinee proficiency generated by $\mathcal{M}$ to obtain the hash representation of student ability.

4.1 Hash Representation Generation

In order to generate suitable input vectors for hash representation, we characterize the embedding features of both examinees and exercises from the perspective of knowledge concepts. Based on the existing examinee practice records $\mathcal{R}$, we encode examinees, exercises, and knowledge concepts through a trainable parameter matrix to obtain the initial embedding vectors. Using these initialized embeddings, we derive meaningful examinee and exercise representation vectors. Specifically, we obtain the examinee's knowledge concept mastery vector $\boldsymbol{\gamma} \in \mathbb{R}^{1\times|C|}$, the exercise's knowledge concept difficulty vector $\boldsymbol{\delta} \in \mathbb{R}^{1\times|C|}$, and the exercise's discrimination factor $\psi \in (0,1)$, as show in Eq (2):

$$\boldsymbol{\gamma} = \boldsymbol{h}^s \times (\boldsymbol{h}_K)^T, \boldsymbol{\delta} = \boldsymbol{h}^e \times (\boldsymbol{h}_K)^T, \psi = \phi(\boldsymbol{h}^e \times \boldsymbol{W} + b), \tag{2}$$

where $\boldsymbol{h}^s = \boldsymbol{x}^s \times \boldsymbol{W}_s, \boldsymbol{h}^e = \boldsymbol{x}^e \times \boldsymbol{W}_e, \boldsymbol{h}^k = \boldsymbol{x}^k \times \boldsymbol{W}_k, \boldsymbol{h}^s, \boldsymbol{h}^e, \boldsymbol{h}^k \in \mathbb{R}^{1\times d}$ represent the initial embedding representations of examinee s, exercise e, and knowledge concept k, respectively. Each knowledge concept is encoded as a d-dimensional vector, allowing us to derive the representation matrix $\boldsymbol{h_K} \in \mathbb{R}^{|C|\times d}$ for the knowledge concept set C. $\boldsymbol{x}^s \in \{0,1\}^{1\times|\mathcal{S}|}$, $\boldsymbol{x}^e \in \{0,1\}^{1\times|\mathcal{E}|}$, $\boldsymbol{x}^k \in \{0,1\}^{1\times|C|}$ are defined as one-hot representation vectors representing examinee s, exercise e, and knowledge concepts k, respectively. $\boldsymbol{W_s} \in \mathbb{R}^{|\mathcal{S}|\times d}, \boldsymbol{W_e} \in \mathbb{R}^{|\mathcal{E}|\times d}, \boldsymbol{W_k} \in \mathbb{R}^{|C|\times d}, \boldsymbol{W} \in \mathbb{R}^{d\times 1}$, are trainable matrices. ϕ denotes the activation function.

After obtaining the knowledge concepts mastery vector γ of the examinee s and the knowledge concepts difficulty vector δ of the exercise e, we generate the corresponding hash representation vectors from these two real-valued representation vectors. To illustrate this process, we use the exercise side as an example.

Specifically, we construct a bidirectional information encoding mechanism to embed input vectors into a low-dimensional latent space with constrained distributions. This generation process comprises two key steps: first, a parameterized encoder maps the input into the statistical parameters (e.g., mean and standard deviation) of a latent distribution; second, latent variables are sampled from this distribution by introducing stochastic perturbations. These latent variables are subsequently transformed into hash representations. To generate binary hash representations (i.e., 0 and 1) while preserving the ability for gradient backpropagation [20], we design a smoothing transformation mechanism based on sampling theory. Specifically, to enhance the sparsity and discriminative power of the hash representation, we extend the latent variables by introducing zero vectors in additional dimensions. This operation provides a clear mapping between each category and its potential hash values. Subsequently, random perturbations sampled from a specific noise distribution are added to the extended vectors, followed by normalization through a parameterized transformation function. This normalization process ensures the generated hash representations align with the desired binary format while maintaining differentiability for optimization, as shown in Eq (3):

$$\boldsymbol{z^e} = \mu + \varepsilon \cdot \sigma, \boldsymbol{b^e} = \frac{\exp\left((\log \boldsymbol{z^e} + g)/\tau\right)}{\sum_j \exp\left((\log \boldsymbol{z_j^e} + g)/\tau\right)}, \tag{3}$$

where μ and σ represent the mean and the standard deviation of the underlying distribution, and ε is a random variable sampled from the standard normal distribution, $\boldsymbol{b^e} \in \{0,1\}^{|C|}$ denotes the hash representation of knowledge concepts difficulty vector, g is the perturbation sampled from the noise distribution, and τ is a smoothing parameter that controls the transformation's smoothness. Smaller values of τ yield hash representations closer to discrete values, while larger values preserve smoothness and differentiability [12].

Ultimately, the continuous sampling mechanism generates the final hash representations, ensuring differentiability during training while producing discrete hash values during inference.

To enhance the generalization capability of the model's latent variables, we incorporate an constraint between the latent distribution and the standard normal distribution into the optimization objective. Consequently, the final optimization objective consists of two components: the reconstruction loss, which measures the discrepancy between δ and its reconstruction $\hat{\delta}$, and the distribution alignment loss, as shown in Eq (4):

$$\mathcal{L}_E^H = \|\boldsymbol{\delta} - \hat{\boldsymbol{\delta}}\|_2^2 + \mathbb{D}_{\mathrm{KL}}\left(q(z|x)\|p(z)\right) \tag{4}$$

4.2 Information Alignment

It is worth noting that the process of converting real-valued representation to hash representation has long been proven to cause quantization loss, resulting in a large difference between the two representations [48]. To ensure the maximum consistency between semantic information and structural information, we designed an information alignment module for real-valued and hash representations. For clarity, we use the exercise side as an example.

To efficiently aggregate neighbor information from both real-valued and hash representations, we employ Locality-Sensitive Hashing (LSH). LSH is widely recognized as an effective method for approximate nearest neighbor search due to its data independence and efficient time complexity, making it particularly suitable for processing large-scale datasets [13]. Considering the continuous nature of real-valued representations and the discrete characteristics of hash representations, we design distinct hash functions to generate hash codes for each representation [33]. Following the principle of LSH [5], examinees or exercises with identical hash codes are projected into the same bucket and treated as a cluster. For the real-valued representation, the hash function we employ is defined as follows:

$$h^e = \left\lfloor \frac{\mathbf{z}^e \cdot \mathbf{a} + \mathbf{c}}{w} \right\rfloor \quad , \tag{5}$$

where $\mathbf{a}$ is a vector randomly sampled from a standard normal distribution, $\mathbf{c}$ is a random number, w is a constant. For the hash representation, we select d dimensions from the existing C-dimensional hash representation to generate the hash code: $h^{e\prime} = b^e[b]$.

From the real-valued data buckets, we can get the similarity matrix $\boldsymbol{A} \in \{1,0\}^{|\mathcal{E}|\times|\mathcal{E}|}$ to describe the data relationship in the bucket. Similarly, we can also get the hash-represented similarity matrix $\boldsymbol{A'} \in \{1,0\}^{|\mathcal{E}|\times|\mathcal{E}|}$. Take real-valued representation as an example in Eq (6):

$$\boldsymbol{A_{i,j}} = \begin{cases} 1, & \text{if } h_i^e = h_j^e \\ 0, & \text{if } h_i^e \neq h_j^e \end{cases} \tag{6}$$

Based on the data we obtained above, combined with the examinee side and the exercise side, We need to optimize the information alignment module in Eq (7):

$$\begin{aligned} \mathcal{L}_{\text{con}} = \mathbb{E}_{i,j}\left[\|z_i^s - z_j^s\|^2 A'_{i,j}\right] + \mathbb{E}_{i,j}\left[(b_i^s)^T b_j^s A_{i,j}\right] \\ + \mathbb{E}_{i,j}\left[\|z_i^e - z_j^e\|^2 A'_{i,j}\right] + \mathbb{E}_{i,j}\left[(b_i^e)^T b_j^e A_{i,j}\right], \end{aligned} \tag{7}$$

where $\|z_i^e - z_j^e\|^2$ denotes the squared Euclidean distance between two real-valued representations vectors. The optimization idea is: taking the exercise side as an example, the goal of the first item is to make the Euclidean distance between real-valued representations as small as possible under similar hash representations; the goal of the second item is to make the hash representation have a larger value on the inner product under similar real-valued representations. Through this optimization method, the consistency loss promotes the consistency between the real-valued representation and the hash representation during training, ensuring that the representation learned by the model effectively captures the potential relationship between examinees and exercises.

4.3 Hash-based Diagnosis

Traditional hash representation methods often rely on inner product or cosine similarity operations to optimize the hash representation generation process [49]. However, in the educational context, such operations fail to capture the intricate relationships between examinees and exercises. To address this limitation and better model the complex interactive relationships, we propose a cognitive diagnosis function tailored for hash representations, as shown in Eq (8):

$$F_H(\boldsymbol{b^s}, \boldsymbol{b^e}) = Q \cdot [\alpha \cdot (1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e}) + \beta \cdot \text{Interaction}(\boldsymbol{b^s}, \boldsymbol{b^e})] \cdot \psi, \tag{8}$$

where Q is the $Q-$matrix describing the relationship between the exercise and the knowledge concepts in the diagnostic task. $\boldsymbol{b^s}$ and $\boldsymbol{b^e}$ represent the binary representations of the examinee s and the exercise e, respectively. ψ is the discrimination of the difficulty of the exercise, $\oplus$ is the XOR operation in the binary operation, Interaction$(\boldsymbol{b^s}, \boldsymbol{b^e})$ is the inner product operation, and α, β are hyper-parameters. Specifically, $\boldsymbol{b^s} \oplus \boldsymbol{b^e}$ represents the difference in knowledge concepts between examinee s and exercise e under the hash representation. The term $1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e}$ evaluates the examinee's mastery of the exercise. A large difference, as indicated by $\boldsymbol{b^s} \oplus \boldsymbol{b^e}$, suggests a lower probability of the examinee s correctly answering the exercise e, as reflected by the value of $1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e}$. Additionally, inspired by operations used in real-valued representations [29], the dot product between $\boldsymbol{b^s}$ and $\boldsymbol{b^e}$ captures the correlation between the examinee's mastery of the knowledge concepts and the exercise e. A higher dot product value indicates a better alignment between the current ability of examinee s and the difficulty of the exercise e, thereby increasing the likelihood of a correct response. It is worth noting that in the educational scenario, the monotonicity assumption is a non-negligible property and this assumption should be transformed into a property of the interaction function [32].

Proof: We first define the function $f(\boldsymbol{b^s})$, expressed as: $f(\boldsymbol{b^s}) = \alpha \cdot (1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e}) + \beta \cdot \text{Interaction}\ (\boldsymbol{b^s}, \boldsymbol{b^e})$. In this formulation, $b^s \oplus b^e$ is used to measure the difference between two hash representation vectors: $XOR_Sum(\boldsymbol{b^s}, \boldsymbol{b^e}) = \sum_{k=1}^{d}(\boldsymbol{b^s}[k] \oplus \boldsymbol{b^e}[k])$, where d is the dimensionality of the hash vectors. When $\boldsymbol{b^s} \oplus \boldsymbol{b^e}$ is more similar, $XOR_Sum(\boldsymbol{b^s}, \boldsymbol{b^e})$ decreases, which leads to an increase in $1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e}$. Consequently, the matching degree between $\boldsymbol{b^s}$ and $\boldsymbol{b^e}$ increases monotonically. Additionally, the term Interaction$(\boldsymbol{b^s}, \boldsymbol{b^e})$, represented by the dot product operation, evaluates the matching degree between the examinee and the exercise. When $\boldsymbol{b^s}$ and $\boldsymbol{b^e}$ become more similar, this term also increases monotonically. To sum up, when function $f(\boldsymbol{b^s})$ takes the partial derivative of $\boldsymbol{b^s}$:

$$\frac{\partial(1 - \boldsymbol{b^s} \oplus \boldsymbol{b^e})}{\partial \boldsymbol{b^s}} \geq 0, \frac{\partial(\text{Interaction}(\boldsymbol{b^s}, \boldsymbol{b^e}) \oplus \boldsymbol{b^e}}{\partial \boldsymbol{b^s}} = \boldsymbol{b^e} \geq 0. \tag{9}$$

Therefore, the total derivative is:

$$\frac{\partial f(\boldsymbol{b}^s)}{\partial \boldsymbol{b}^s} = \alpha \cdot \frac{\partial (1 - \boldsymbol{b}^s \oplus \boldsymbol{b}^e)}{\partial \boldsymbol{b}^s} + \beta \cdot \boldsymbol{b}^e \geq 0. \tag{10}$$

Since $\alpha, \beta > 0$, it can be seen that $f(\boldsymbol{b}^s)$ is monotonically increasing for $\boldsymbol{b}^s$, satisfying the monotonicity assumption.

Finally, we use the cross entropy between the output $y_{ij} = F_H(\boldsymbol{b}^s, \boldsymbol{b}^e)$ and the true label r_{ij} to optimize the diagnostic function under the hash representation, as shown in Eq (11):

$$\mathcal{L}_{\text{Hcdm}} = -\sum_{ij} \left(r_{ij} \log y_{ij} + (1 - r_{ij}) \log(1 - y_{ij}) \right). \tag{11}$$

In summary, the fusion loss function of the entire hash representation generation process is:

$$\mathcal{L}_H = \mathcal{L}_{Hcdm} + \mathcal{L}_S^H + \mathcal{L}_E^H + \mathcal{L}_{con}. \tag{12}$$

4.4 Uncertainty Reduction-based Exercise Selection

After the previous stage, we can get the hash representation of the exercise and the two real-valued information of exercise difficulty and exercise discrimination. In the CAT process, we adopt the same method used for generating exercise hash representations to process the examinee ability information obtained from the base cognitive diagnosis model into hash representations.

In designing the selection strategy, we adopted a method based on uncertainty reduction. The core idea is to select exercises that minimize the uncertainty in assessing a examinee' ability. To quantify this uncertainty, we introduced the concept of information entropy, a widely used metric for measuring the uncertainty of random variables [4]. The larger the entropy value, the higher the uncertainty in the assessment of examinee ability; conversely, the smaller the entropy value, the lower the uncertainty [27]. Specifically, exercises are selected by calculating the reduction in uncertainty (i.e., the change in entropy) that each exercise contributes to the examinee ability assessment:

$$\Delta H = H_{prior} - H_{post}, \tag{13}$$

where H_{prior} is the information entropy before exercise selection, representing the uncertainty in assessing the examinee's ability prior to answering a exercise. H_{post} is the information entropy after exercise selection, representing the uncertainty in assessing the examinee's ability after answering the selected exercise. Our objective is to select exercises that maximize ΔH, thereby prioritizing those that lead to the greatest reduction in uncertainty.

Specifically, based on the current examinee ability and exercise characteristics, the probability P_c of the examinee answering each exercise correctly is predicted, and P_c is used to calculate the current information entropy:

$$P_c = \{1 + exp\{-\kappa \cdot (M_{se}(b^s, b^e) \cdot \psi) + \zeta \cdot \mathbb{E}(\delta)\}\}^{-1}, \tag{14}$$

$$H_{prior} = -P_c \cdot log(P_c) - (1 - P_c) \cdot log(1 - P_c), \tag{15}$$

where M_{se} denotes the operation of 1 minus the Hamming distance between b^s and b^e, $\mathbb{E}(\delta)$ denotes the mean of the exercise difficulty vector δ. After we select a exercise, we estimate the information entropy H_{post} of the examinee after solving the exercise. Here we assume that after the examinee solves a exercise, the uncertainty of the prediction of the examinee's ability decreases, and reduce H_{prior} through certain rules. We define a difficulty adjustment factor combined with exponential decay to measure the change of information entropy, as follows:

$$H_{post} = H_{prior} \cdot (-\frac{\delta_{gap}}{exp(\mathbb{E}(\delta))}), \tag{16}$$

where $\delta_{gap} = |\mathbb{E}(\theta) - \mathbb{E}(\delta)|$, $\mathbb{E}(\theta)$ denotes the mean of the examinee's current abilities, as evaluated during the CAT exercises selection process. By incorporating this metric, we can account for both the abilities of the examinee and the difficulty of the exercise. This ensures the selection of personalized exercises tailored to each examinee's needs.

In addition, during the CAT process, the assessment of examinee ability is progressively updated with each step of exercise selection and response, so the dynamic decision-making process is typically regarded as a Markov process [50]. Thus, we incorporate the ϵ-greedy and UCB strategies into the selection algorithm. We designed a hybrid algorithm that combines two strategies. Specially, during the first S steps of the exercise selection, the ϵ-greedy is applied. In this process, a exercise is randomly selected with a probability of ϵ, while the exercise with largest ΔH is greedily chosen with a probability of $1-\epsilon$. Additionally, the number of times each exercise is selected and its corresponding ΔH are recorded as rewards. In the subsequent S steps, we adopt the UCB(Upper Confidence Bound [28]) strategy to further balance exploration and exploitation. The UCB strategy selects the exercise with the highest UCB value, calculated as the upper confidence bound of each exercise. Specifically, the UCB of the exercise consists of two components, as shown in Eq(17):

$$UCB = \frac{Re}{N + \omega} + \sqrt{\frac{2 \cdot \ln(TQ)}{N + \omega}}, \tag{17}$$

where Re denotes the cumulative reward of the exercise , defined as the sum of ΔH values for all instances where exercise is selected. N denotes the number of times exercise has been selected, while TQ is the total number of selections across all exercises. ω is a constant introduced to prevent division by zero errors.Precisely, the first part, the Exploitation Term, reflects the known performance of the exercise. The second part, the Exploration Term, increases as an exercise is selected less frequently, encouraging the algorithm to explore under-selected exercises. By combining these components, the UCB strategy ensures the selection of exercises that minimize uncertainty (as measured by ΔH) while exploring potential high-value exercises. This approach enables the algorithm to achieve more effective and robust evaluation results over time.

5 Experiments

We conduct experiments on four real-world educational datasets to evaluate our CAT approach, addressing four key questions: **RQ1**: How effectiveness and efficient is our proposed CAT approach in improving the overall performance of the CAT process? **RQ2**: Are the designed key components conducive to the performance improvement of our CAT approach? **RQ3**: What are the underlying mechanisms that enable our CAT approach to enhance exercise

Table 1: The statistics of the datasets.

DataSet	NIPS-2020	JUNYI	ASSIST1213	PTADISC
#Student	3,934	54,564	16,412	18,768
#Question	900	565	3,906	3,262
#Concept	56	30	119	50
#Log	1,222,047	1,711,210	970,202	5,720,582

selection efficiency? **RQ4**: How sensitive is the our CAT approach to parameters?

5.1 Experimental Settings

In this section, we introduce the datasets, the selected baselines , and the application of CAT.

5.1.1 Dataset Description. We used four real-world educational datasets, named NIPS-EDU, ASSIST1213, JUNYI, and PTADISC. NIPS-EDU [35] is a dataset for the NeurIPS 2020 Education Challenge, and it comes from students' answers to exercises in Eedi (an educational platform). ASSIST1213 [22] is collected from the online tutoring system ASSISTments, and we used the dataset between 2012 and 2013. JunYi [3] is also a dataset collected from an online learning website. PTADisc [11] originated from the online learning platform PTA, which is an automated project evaluation and open teaching assistance platform for colleges and society. We used the data processing approach of previous work [50], using 80%-10%-10% of students for training, validation, and testing respectively. The statistics of the processed datasets are shown in Table1. Among them, Data-Driven approaches need to be trained and tested based on the support set and query set.

5.1.2 Baseline approaches. To demonstrate the effectiveness of our approach, we compared it with exercise selection strategies that used two different cognitive diagnosis modules. For the cognitive diagnosis module, we used two representative cognitive diagnosis models: NCD and MIRT. We selected the following state-of-the-art selection strategies as baselines: (1) Random: the random selection algorithm. (2) MKLI [2]: It selects exercises with the largest moving average of Kullback-Leibler information, ensuring efficient evaluation and supporting multidimensional student abilities. (3) MAAT [1]: It's an active learning-based approach including quality module, diversity module, and importance module. It uses Expected Model Change (EMC) of CDM to select exercises. (4) BECAT [51]: It is an approach for estimating the bounded ability of CAT by selecting a subset of exercises that aligns with the gradient of the full answer. (5) NCAT [50]: It defines the CAT problem as a bi-level optimization problem, uses reinforcement learning to solve the CAT task, and proposes an attention neural strategy to model the complex nonlinear interactions between examinees and exercises. (6) GMOCAT [34]: It uses graph neural networks to embed the relationship between exercises and concepts, and introduces three goals: quality, diversity, and novelty in a multi-objective reinforcement learning framework. (7) SECAT [10]: It is a CAT search framework that achieves efficient search in the topic selection process by recursively dividing the entire exercise space into many logarithmic-sized subspaces. Here we apply this framework to MAAT.

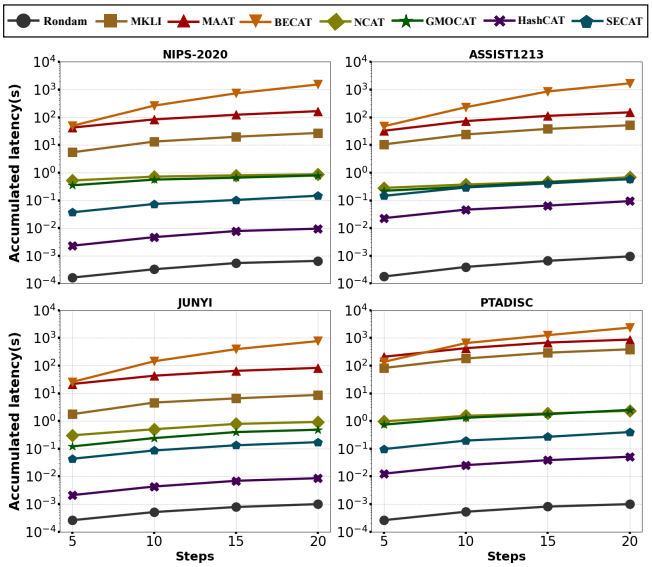


Figure 3: The accumulated latency of our CAT approach and all baseline approaches based on NCD across four datasets.

5.1.3 Evaluation Metrics. We used two types of evaluation metrics: effectiveness and efficiency. Effectiveness: We evaluated the accuracy of the final ability estimate by predicting the students' binary responses to the exercises in the test set. Therefore, we used accuracy (ACC) and area under the ROC curve (AUC) as evaluation metrics [40]. Efficiency: We used the average evaluation time (seconds) per student under a specific test step length to measure the efficiency of exercise selection. Moreover, we compare the training time (seconds) performance of data-driven methods with our proposed hash representation generation module.

5.1.4 Parameter Settings. In the pre-training stage of the cognitive diagnosis model, we set the embedding dimensions of students and exercises in NCD and MIRT to the number of knowledge concepts in the dataset. In addition, we uniformly set the batch size and learning rate to 128 and 0.001 in this stage. In the hash representation generation stage, we set the low-dimensional embedding dimension d of examinees and exercises to 20, w in the hash function of the real-valued representation in the LSH process to 0.5, and α and β in the hash-based diagnosis function to 0.5 and 0.7 respectively. At the same time, in this stage, the batch size and learning rate are set to 1024 and 0.0005 respectively. We initialize all parameters with Xavier and use the Adam optimizer. In the CAT evaluation stage, the exercise selection algorithm follows the settings in the original paper, and the test length is set to 5, 10, and 20. We conduct our experiments on an NVIDIA RTX 4090 GPU by PyTorch.

5.2 Performance Comparison (Q1)

To demonstrate that our HashCAT approach can balance time efficiency and evaluation effectiveness, we compared it with other baseline approaches. Table 2 presents the ACC and AUC metrics after t steps of exercise selection, as well as the average exercise selection time per student when $t = 20$. Figure 3 illustrates the exercise selection time using NCD as the CDM four datasets. In addition, table 3 shows the average training time over 20 iterations for our

Table 2: Performance on four public datasets is evaluated based on ACC, AUC, and the average exercise selection time per student. The best performance is highlighted in bold, the second-best in underlined text, and our method is marked in gray. "-" indicates that the student abilities and exercise features generated by MIRT are not applicable for optimizing the dual-tower model in SECAT, and "*" denotes that the approach takes excessively long to run at step=20 and is therefore not recorded.

Dataset		NIPS-2020													
CDM		MIRT							NCD						
Metric		ACC(%) ↑			AUC(%) ↑			Latency(s) ↓	ACC(%)↑			AUC(%)↑			Latency(s)↓
Step		5	10	20	5	10	20	Total/stu	5	10	20	5	10	20	Total/stu
Statistical	Random	61.59	62.69	63.23	68.26	69.95	70.09	**9.46e-5**	60.67	61.05	63.23	66.24	67.65	69.02	**6.52e-5**
	MKLI	62.33	63.00	64.01	69.39	70.58	71.54	42.028	60.91	62.12	63.88	67.16	67.99	69.80	27.095
	MAAT	63.60	64.05	64.52	69.62	71.12	72.86	162.586	61.69	62.95	65.56	67.46	68.91	71.53	163.957
	BECAT	64.23	65.28	66.24	69.94	71.67	73.40	1610.842	61.52	63.15	66.08	67.52	69.27	72.02	1523.691
Data-Driven	NCAT	64.78	66.09	67.11	69.85	72.05	74.54	0.7398	64.81	66.26	68.66	66.30	67.41	68.40	0.8702
	GMOCAT	**65.32**	66.68	68.45	69.20	72.26	74.95	0.6738	**64.94**	**66.88**	**69.03**	**68.72**	**70.24**	70.73	0.7933
SubSpace	SECAT	–	–	–	–	–	–	–	61.14	62.25	64.45	67.53	69.94	70.54	0.1468
Hash	HashCAT	64.86	**66.90**	**69.03**	**70.03**	**72.62**	**75.62**	0.0093	61.84	63.22	66.21	68.22	70.07	**73.08**	0.0095
Dataset		JUNYI													
Statistical	Random	85.17	85.94	86.01	82.56	83.10	84.01	**8.76e-5**	59.01	61.09	62.52	76.69	77.90	79.04	**9.95e-5**
	MKLI	86.38	86.67	86.80	82.56	83.86	85.45	15.960	59.71	61.80	64.15	77.83	78.92	82.10	8.721
	MAAT	86.34	86.89	86.99	84.14	85.49	86.13	80.261	60.40	62.04	66.67	78.01	78.58	79.88	82.654
	BECAT	86.62	86.78	87.20	84.98	85.73	86.89	761.563	60.54	62.05	66.33	77.29	78.78	80.41	764.648
Data-Driven	NCAT	86.34	86.95	87.35	86.27	86.42	87.17	0.8310	62.83	63.74	68.53	77.67	78.84	80.97	0.9235
	GMOCAT	86.62	87.24	87.77	**86.54**	87.07	87.79	0.2882	**63.03**	**64.18**	**70.37**	77.82	79.04	81.26	0.4837
SubSpace	SECAT	–	–	–	–	–	–	–	60.46	61.94	65.13	77.47	78.48	80.50	0.1719
Hash	HashCAT	**87.44**	**87.61**	**87.80**	86.24	**87.27**	**87.85**	0.0082	60.58	62.61	68.48	**78.53**	**79.84**	**82.16**	0.0086
Dataset		ASSIST1213													
Statistical	Random	70.42	72.05	73.10	69.88	71.05	72.69	**1.002e-4**	65.76	66.37	66.90	69.39	69.89	70.27	**6.52e-5**
	MKLI	70.89	72.62	72.73	70.92	72.53	73.32	67.684	67.32	67.46	67.92	70.84	71.31	71.95	51.442
	MAAT	72.52	72.87	73.07	71.24	73.92	75.86	121.470	67.51	67.87	68.49	70.43	70.53	71.66	150.189
	BECAT	72.60	73.17	74.18	71.30	73.13	75.89	1633.142	67.65	67.98	68.79	70.54	70.71	71.45	1673.237
Data-Driven	NCAT	72.20	73.28	74.67	71.63	73.25	74.18	0.5487	68.27	69.40	70.90	71.08	71.60	73.43	0.6860
	GMOCAT	**72.74**	**73.74**	75.10	**71.76**	**74.04**	75.98	0.5013	**68.78**	**69.98**	**71.23**	**71.45**	**72.03**	**73.98**	0.5767
SubSpace	SECAT	–	–	–	–	–	–	–	67.52	67.82	69.03	70.72	70.82	71.03	0.5877
Hash	HashCAT	72.62	73.51	**75.16**	71.35	73.98	**76.73**	0.0892	67.69	68.27	69.36	70.86	71.37	72.23	0.0936
Dataset		PTADISC													
Statistical	Random	70.18	70.99	71.10	67.01	67.87	68.65	**9.46e-5**	59.45	60.10	60.63	66.48	67.10	68.01	**9.93e-5**
	MKLI	73.50	73.89	74.25	69.10	69.55	70.23	393.435	60.05	60.33	60.88	67.43	67.59	68.05	386.382
	MAAT	73.73	74.01	*	69.17	69.64	*	*	61.08	61.78	62.20	68.15	68.70	69.03	867.421
	BECAT	73.87	74.20	*	69.31	69.70	*	*	61.37	61.88	62.34	68.39	68.76	69.10	2345.132
Data-Driven	NCAT	73.91	74.37	75.27	69.40	69.82	70.85	2.3019	62.13	63.80	64.10	69.38	70.46	71.34	2.3287
	GMOCAT	**76.09**	**76.88**	**77.52**	**70.31**	**71.67**	**72.25**	2.2954	**62.29**	**64.07**	**65.72**	**70.62**	**71.19**	**71.95**	2.5044
SubSpace	SECAT	–	–	–	–	–	–	–	61.20	62.28	62.45	68.50	68.64	69.05	0.3967
Hash	HashCAT	74.01	74.50	75.55	69.42	69.73	71.49	0.0532	61.55	63.03	64.02	68.48	68.82	69.16	0.0511

approach compared to two data-driven approaches. We have the following observations: (1) Our approach surpasses most CAT methods in evaluation effectiveness and ranks second only to random selection in time efficiency, showcasing robust performance. This demonstrates that the uncertainty reduction-based exercise selection strategy, utilizing hash representations, successfully balances evaluation accuracy and computational efficiency. (2) Compared to data-driven approaches, our method demonstrates notable advantages in training efficiency, particularly when NCD is employed as the CDM. While data-driven methods achieve superior evaluation effectiveness on NCD, their reliance on reinforcement learning for exercise selection—requiring extensive agent-environment interactions and complex optimization processes—leads to significantly higher training costs. When MIRT is used as the CDM, our approach achieves comparable performance to data-driven methods. In contrast, statistical methods exhibit lower time efficiency due to their higher computational complexity, and our approach also surpasses them in evaluation performance. (3) In terms of selection time, data-driven approaches exhibit similar efficiency, likely because the trained agent operates more effectively during the evaluation process. BECAT initially demonstrates time efficiency comparable to other statistical methods; however, as the number of steps increases, its time cost rises significantly. This may be attributed to its specific subset selection strategy, which becomes computationally demanding over time.

5.3 Ablation Study (RQ2)

In this section, we perform ablation studies on two datasets, NIPS-2020 and ASSIST1213, using NCD to analyze the impact of key components in HashCAT. To achieve this, we consider three variants of HashCAT, each created by removing a specific component from the original model. The details of these variants are as follows: (1) w/o-IAM, which excludes the Information Alignment Module (IAM) described in Section 4.2. (2) w/o-UR, which excludes the uncertainty reduction strategy described in Section 4.4. The selection criterion is solely based on choosing the problem with the largest P_c.

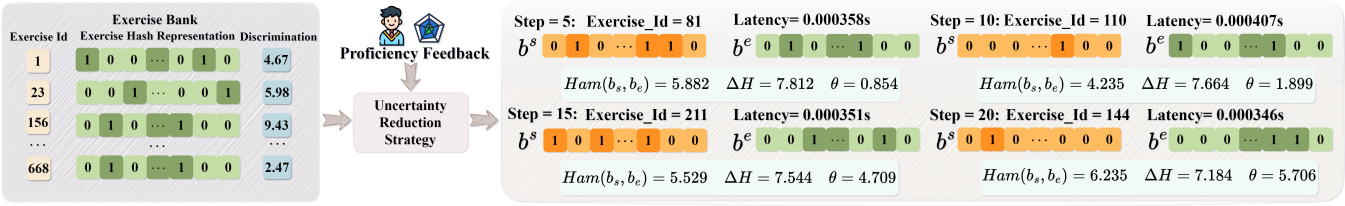


Figure 4: Visualization of exercise selection method. On the far left of this image is a exercise bank, which includes the hash representations and actual value information for each exercise. On the far right are some important parameters during the exercise selection process at steps 5, 10, 15 and 20.

Table 3: Training Time Comparison (Seconds).

Datasets	NIPS-2020	JUNYI	ASSIST1213	PTADISC
NCAT	60.979	57.339	171.193	624.324
GMOCAT	160.007	155.425	274.679	1411.5769
HashCAT	44.477	39.207	137.605	447.037

Table 4: Ablation study on key components.

DataSets	NIPS-2020			ASSIST1213		
Metric	ACC(%) ↑/ AUC(%) ↑			ACC(%) ↑/ AUC(%) ↑		
Step	5	15	20	5	15	20
w/o-IAM	61.35/68.00	64.13/71.27	65.20/72.05	67.41/70.61	67.98/71.10	68.14/71.32
w/o-UR	63.15/62.93	63.63/63.58	63.77/63.97	68.13/67.02	68.55/67.87	68.76/68.15
w/o-Hyb	61.46/67.97	64.48/71.58	66.09/72.87	67.67/70.56	68.18/71.14	68.84/71.63
HashCAT	61.48/68.25	64.86/71.73	66.23/73.09	67.72/70.87	68.86/71.81	69.37/72.24

(3) w/o-Hybrid, which removes the hybrid algorithm component outlined in Section 4.4.

From Table (4), several key findings can be drawn. First, the complete model achieves the best overall performance. Second, when the proposed uncertainty reduction strategy is removed, the performance significantly drops, indicating that relying solely on the Hamming distance between hash representations as a selection strategy does not guarantee accurate assessment results. Third, when the information alignment module or the mixed strategy is removed, performance also declines to some extent. This not only highlights the critical role of the information contained in the hash representations for the assessment results, but also suggests that in the context of a Markov process like CAT, fully exploring and utilizing the untested exercise pool contributes to more accurate assessment of student abilities. Therefore, all components influence HashCAT to varying degrees.

5.4 Case Study and Analysis (Q3)

To further illustrate the details of our exercise selection approach, we have visualized several key metrics of the selection process on the dataset NIPS-2020, in the Figure 4. From the figure, it is evident that during the exercise selection process, student ability progressively improves with each selection step, while the Hamming distance generally exhibits an increasing trend, and the change in information entropy gradually decreases. This is because, as the test progresses, the selection strategy shifts from "adapting to the student's current ability" to "challenging the ability boundary," leading to an increase in Hamming distance. Meanwhile, the reduction in the change of information entropy reflects the gradual convergence of ability estimation uncertainty. We can also observe that the time required for selecting an exercise is remarkably short, primarily because our approach leverages the computational advantages of hash representations in bitwise operations. During the selection process, we directly obtain the hash representations of the exercises and their associated features through the hash generation module. This allows us to use these pre-stored feature data directly during selection, avoiding the bottleneck of real-time data computation found in traditional statistical-based selection algorithms. This demonstrates that our method achieves optimization between student ability assessment and evaluation efficiency, thereby validating the effectiveness of the exercise selection strategy based on hash representations and uncertainty reduction.

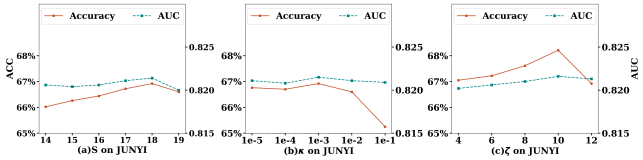


Figure 5: Hyper-parameter sensitivity analysis.

5.5 Effect of Hyperparameters (Q4)

We use NCD in Figure 5 to analyze the impact of S, κ, ζ on the experimental results using the JUNYI dataset. S is a threshold that mixes the ϵ-greedy strategy and the UCB strategy in the exercise selection algorithm, while κ and ζ are parameters used to calculate the probability of a student correctly answering an exercise in H_{prior}. It is observed that the different hyperparameters lead to different results. Specifically, as show in Figure 5 (a), when S equals 18, the approach achieves the best performance on the JUNYI dataset. With ζ fixed at 10, Figure 5 (b) shows that the model performs best when κ equals 0.001. When κ is fixed at 0.001, Figure 5 (c) shows that the model performs optimally when ζ equals 10.

6 Conclusion

This paper presents a CAT method based on hash learning and uncertainty reduction, designed to balance evaluation effectiveness and time efficiency. By combining a hash representation generation module and an uncertainty reduction-based exercise selection algorithm, we successfully integrated hash representations into CAT, significantly enhancing time efficiency. Experimental results on four real-world datasets show that our method not only outperforms most existing approaches in assessment accuracy but also achieves time efficiency comparable to random selection strategies.

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